

4. Write a C program Using recursion solve the Tower of Hanoi problem.

```
#include <stdio.h>
```

```
// Function Prototype
```

```
void tower(int, char, char, char);
```

```
int main(void)
```

```
{
```

```
    int n;
```

```
    printf("Enter the number of disks: ");
```

```
    scanf("%d", &n);
```

```
    printf("The sequence of moves involved in the Tower of Hanoi are :\n");
```

```
    // Function call: n disks, Source='A', Destination='C', Auxiliary='B'
```

```
    tower(n, 'A', 'C', 'B');
```

```
    return 0;
```

```
}
```

```
// Recursive function definition
```

```
void tower(int n, char from_rod, char to_rod, char aux_rod)
```

```
{
```

```
    if (n == 1)
```

```
    {
```

```
        // Base case: move the only disk from source to destination
```

```
        printf("Move disk 1 from rod %c to rod %c\n", from_rod, to_rod);
```

```
        return;
```

```
    }
```

```
    // Step 1: Move n-1 disks from source to auxiliary rod
```

```
    tower(n - 1, from_rod, aux_rod, to_rod);
```

```
    // Step 2: Move the nth disk from source to destination rod
```

```
    printf("Move disk %d from rod %c to rod %c\n", n, from_rod, to_rod);
```

```
    // Step 3: Move the n-1 disks from auxiliary rod to destination rod
```

```
tower(n - 1, aux_rod, to_rod, from_rod);  
}
```